Feature demo · issue #167

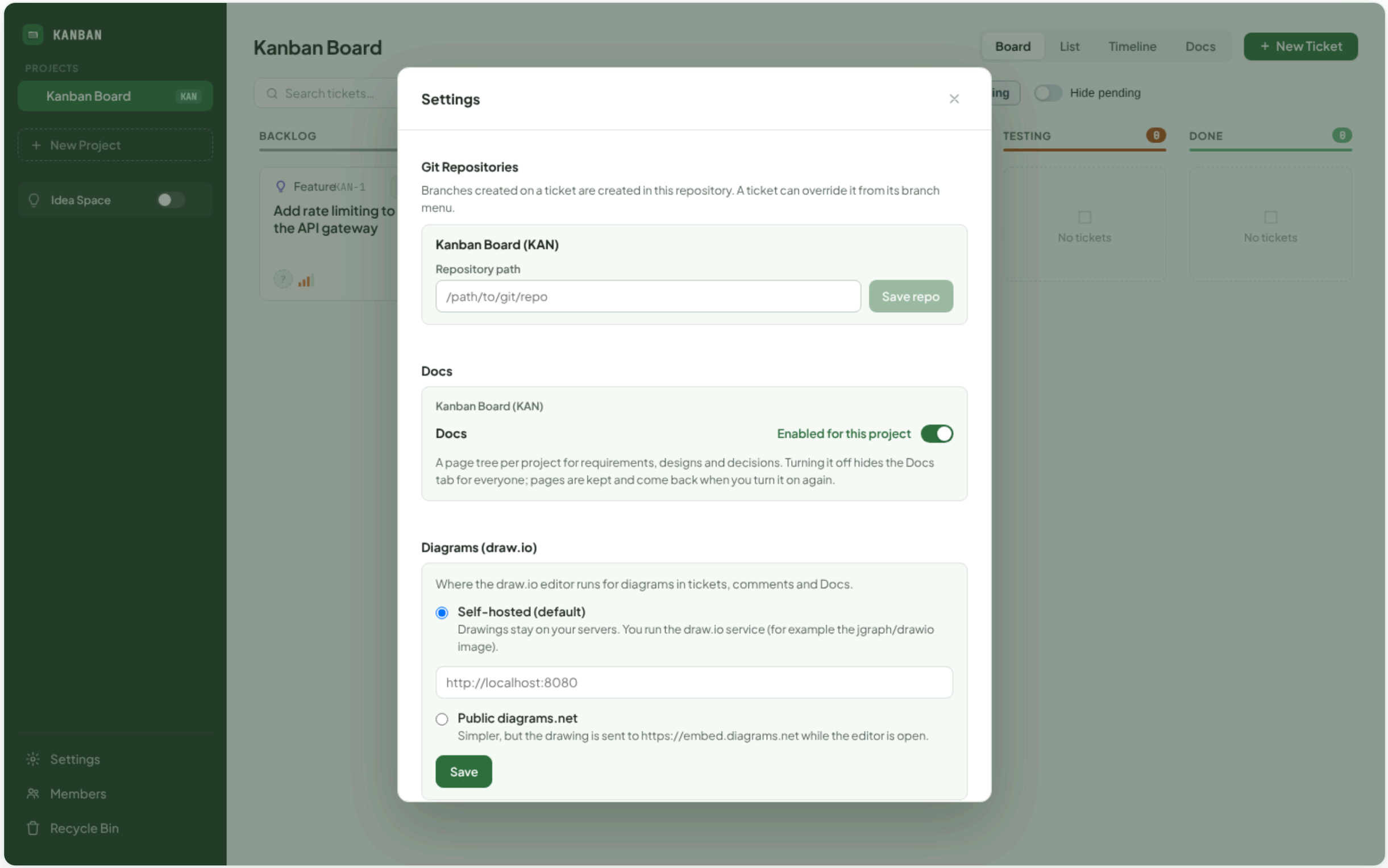
Diagrams — draw.io

Draw flows and sketches inside any markdown field — recorded from the real app (server + UI + draw.io editor), driven by Playwright

- **Where it works**: ticket description, ticket comments and Docs pages. One markdown line per diagram: `[Auth flow](diagram:dg_8f2a)`.
- **Storage**: editable `.drawio` source + an SVG preview made on save; download as .drawio, SVG or PNG 2x; 2 MB limit with a clear error.
- **Editing**: the real draw.io editor in a full-screen layer with our header bar; local draft; confirm before discarding; edit-conflict detection (Save as copy / Reload theirs).
- **Docs history**: each page version keeps the diagram versions it used — restoring a page restores its diagrams. Deleting from the text only unlinks (kept for the Recycle Bin).
- **Permissions**: members add and edit; viewers view and download.
- **Decision**: the editor host is an admin setting, **self-hosted by default**, public diagrams.net optional.

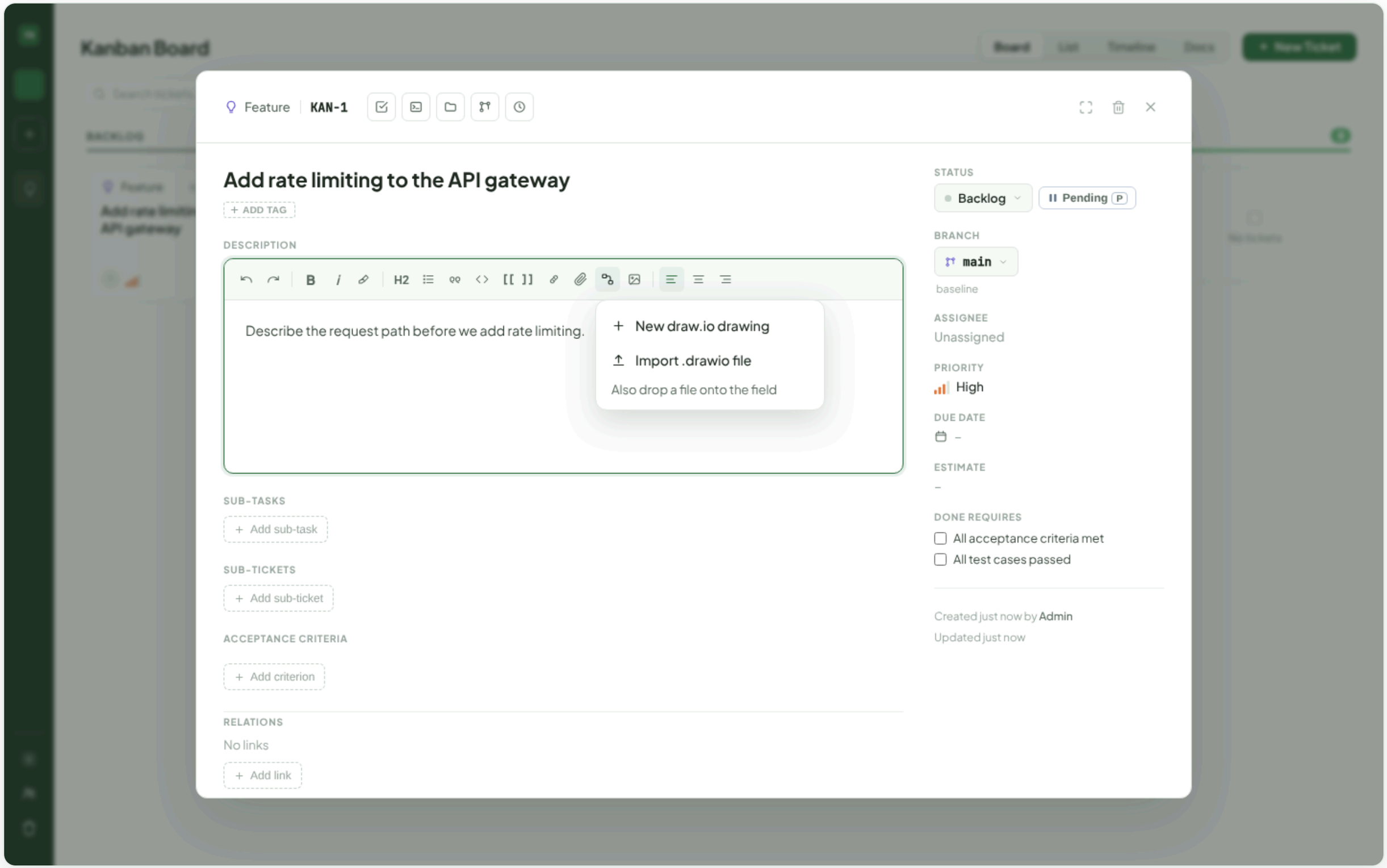
1. Where the draw.io editor runs

Settings › Diagrams (draw.io). Self-hosted is the default (drawings stay on your servers); the public diagrams.net editor is an admin choice and the screen says the drawing is sent there while the editor is open.



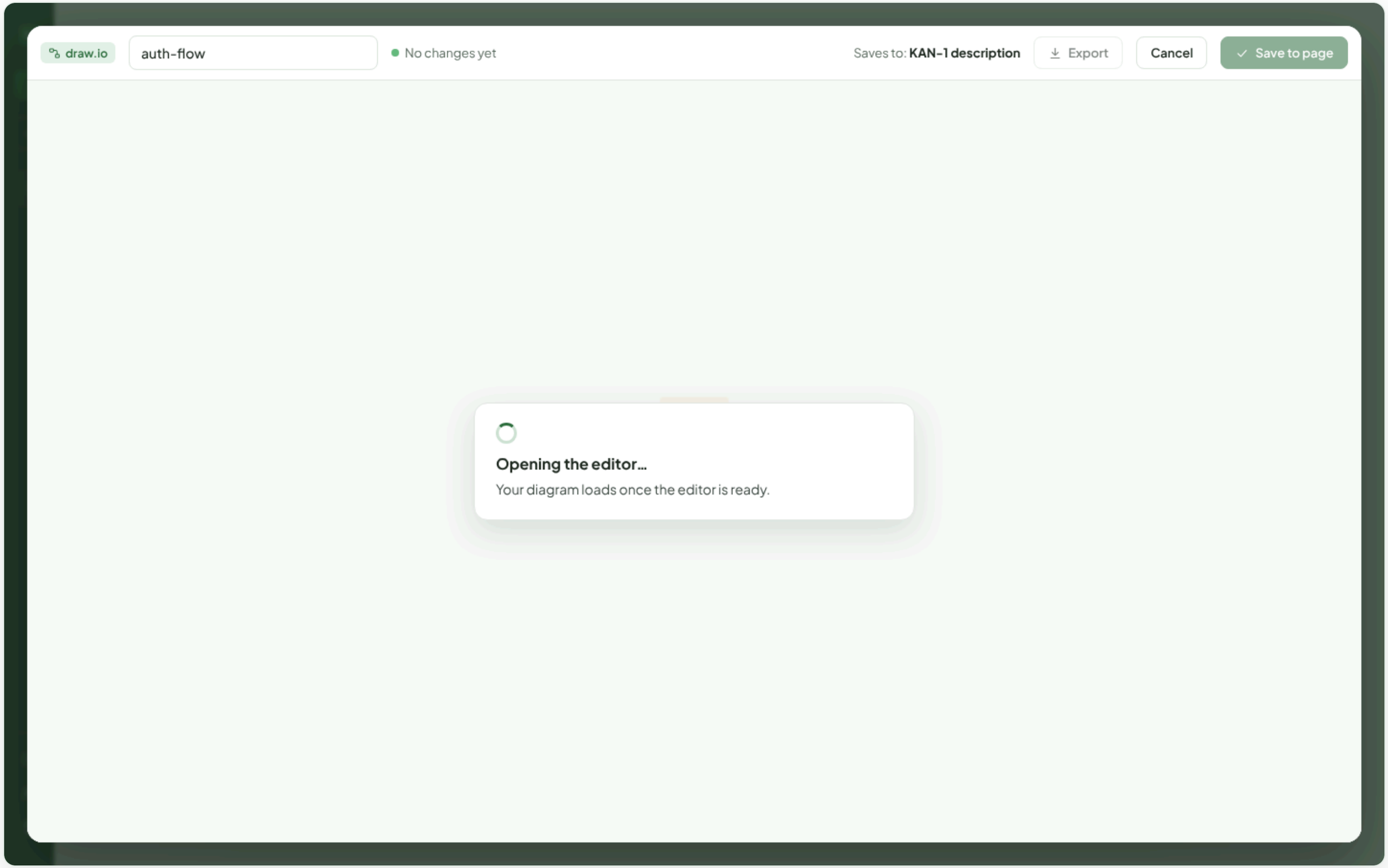
2. Start a drawing from the description toolbar

The Diagram action sits in the markdown toolbar of the ticket description: New draw.io drawing, or Import a .drawio / .xml / .svg file (choose it, or drop it onto the field). The comment composer has the same action as an icon.



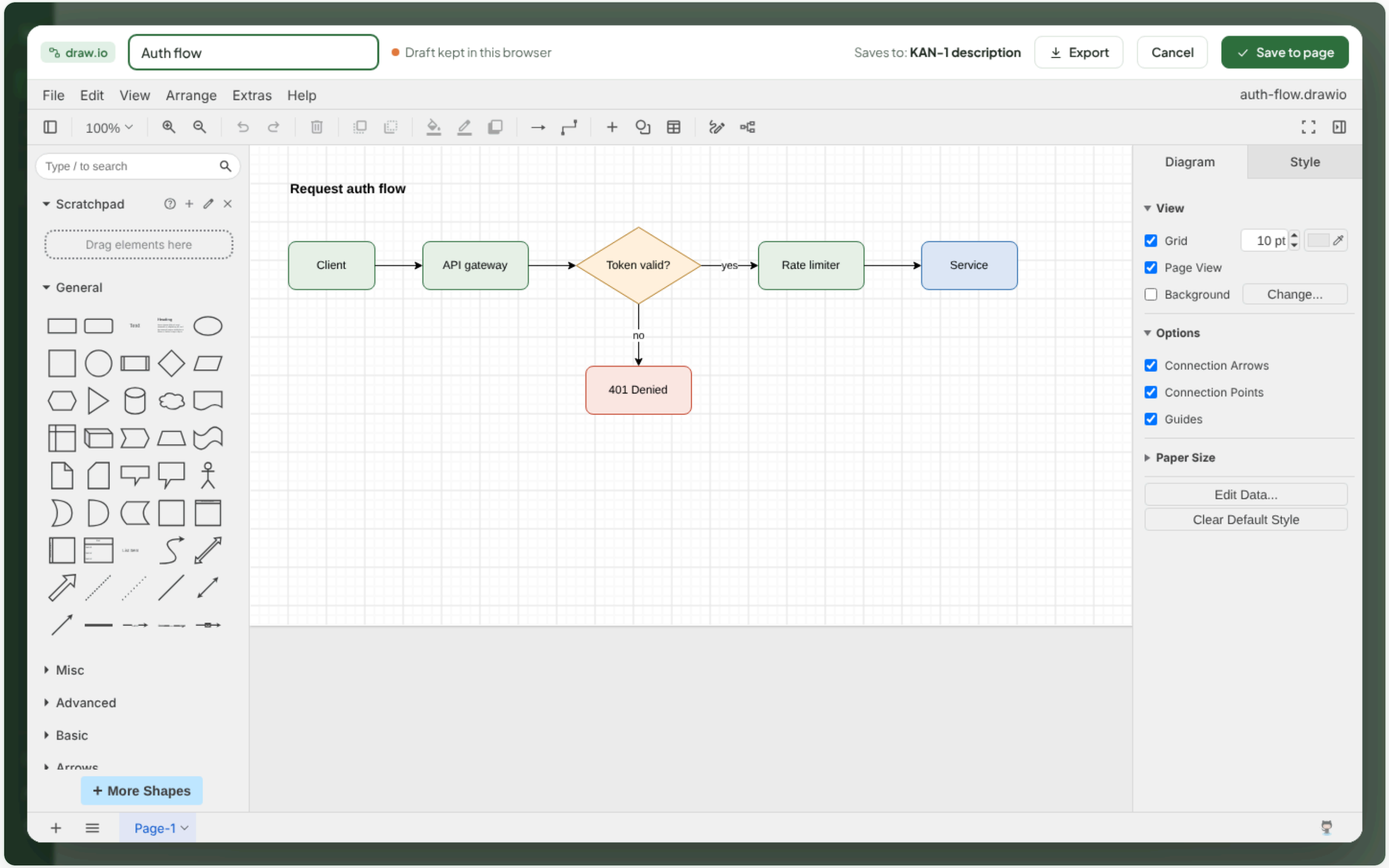
3. The editor layer opens: loading state

A full-screen layer over the page with our own header bar (title, draft status, Saves to, Export, Cancel, Save to page). The imported file is loaded into the editor so it can be checked before saving.



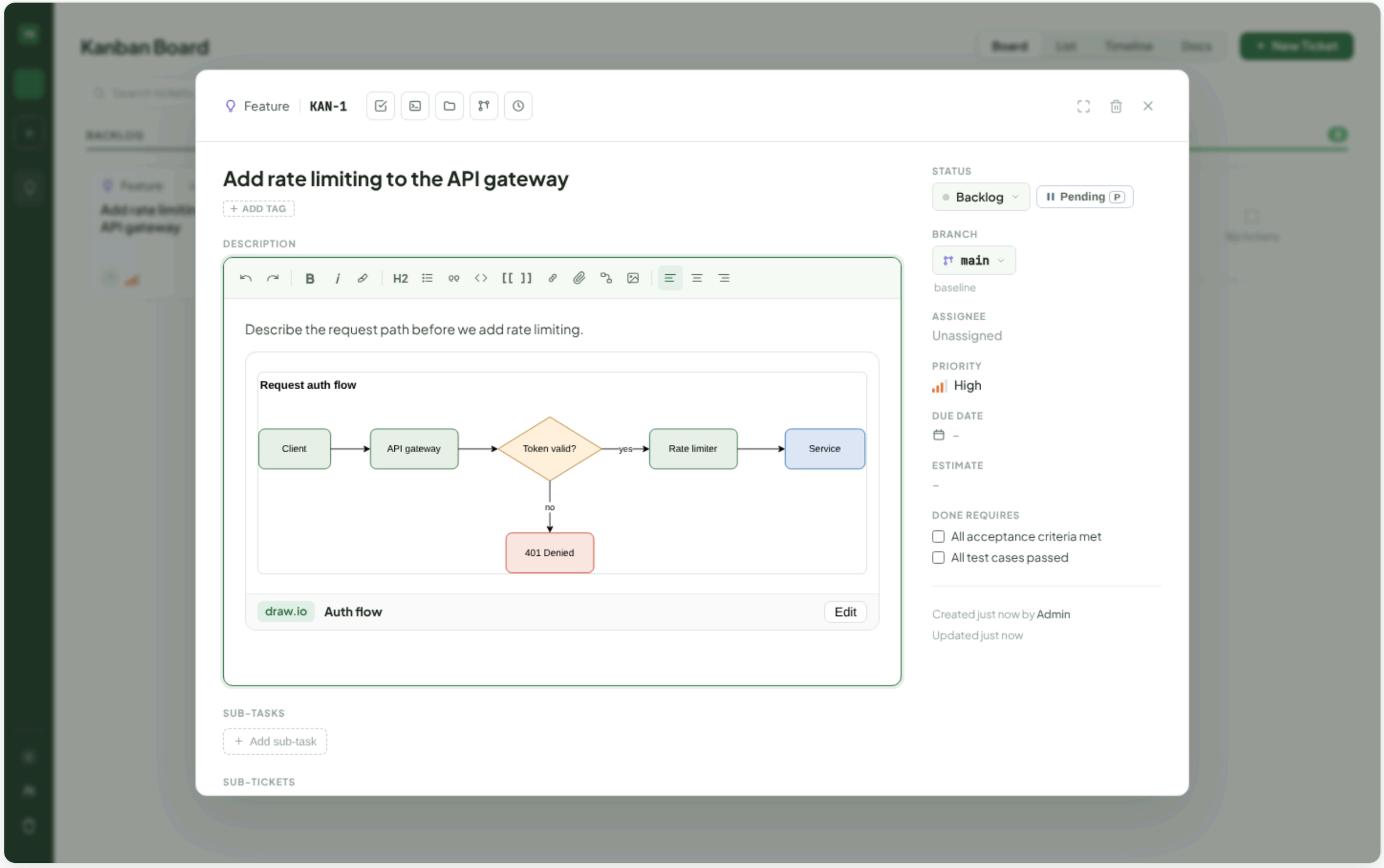
4. The real draw.io editor

Shape library, canvas and format panel are draw.io itself (the self-hosted webapp, driven over its embed protocol). Only the header bar is ours.



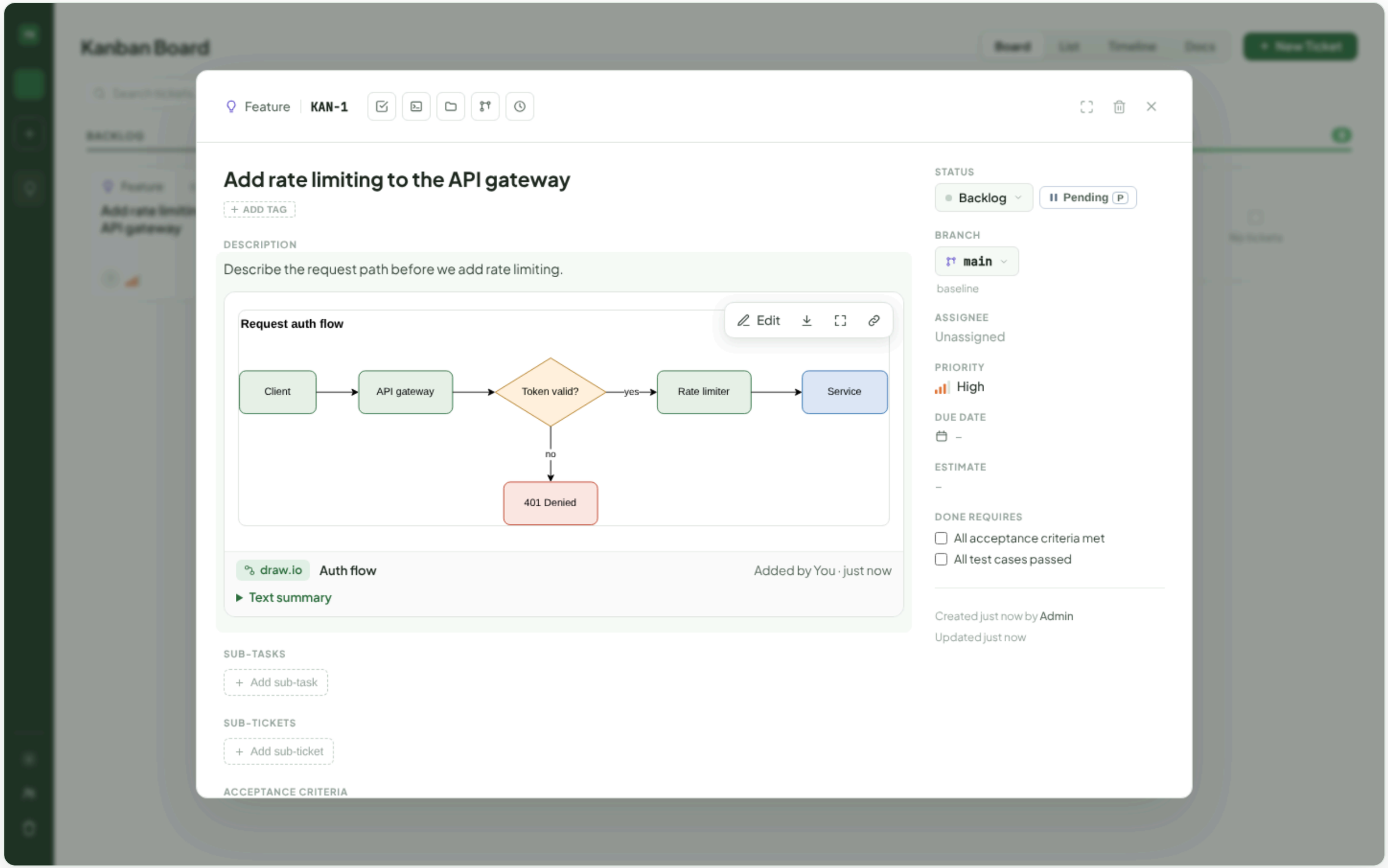
5. Save to page: the diagram lands in the description

The line [Auth flow](diagram:dg_...) is added to the markdown; in the editor it shows as the drawing with an Edit button.



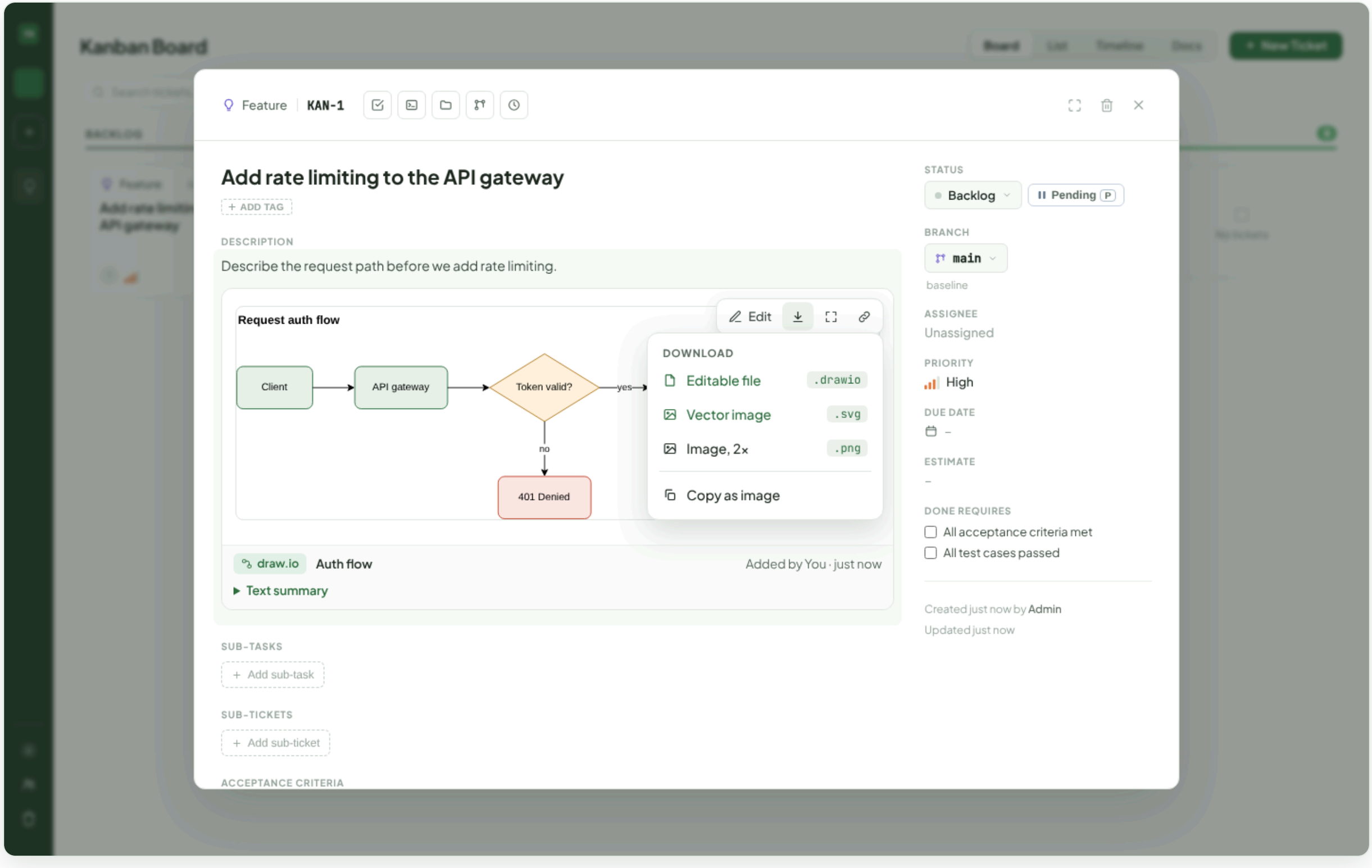
6. How a diagram looks once inserted

A crisp SVG preview, a hover / focus toolbar (Edit, Download, View full size, Copy link), the name, who edited it and when, and a Text summary drafted from the labels.



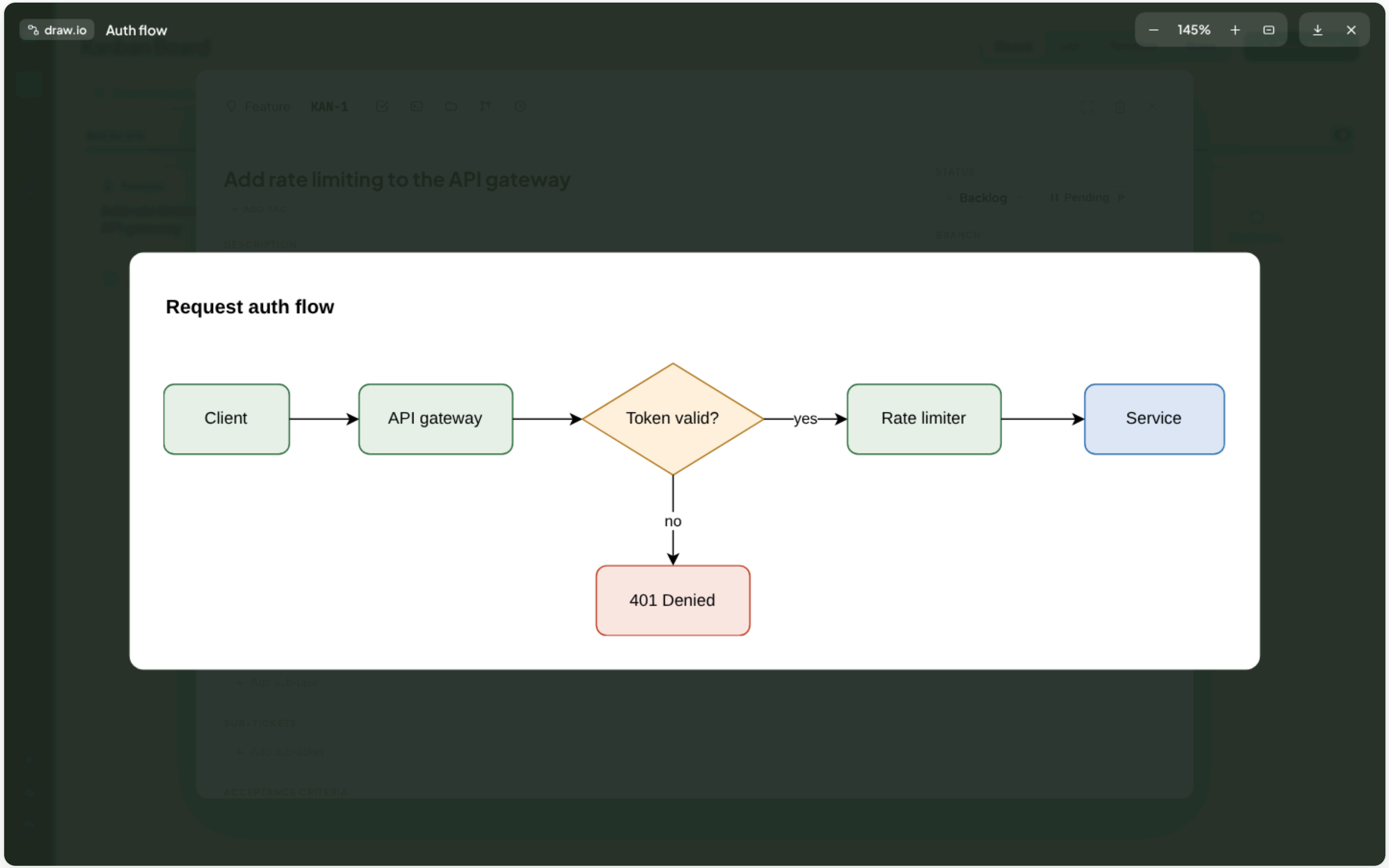
7. Download menu

Editable file (.drawio), Vector image (.svg), Image 2x (.png) and Copy as image.



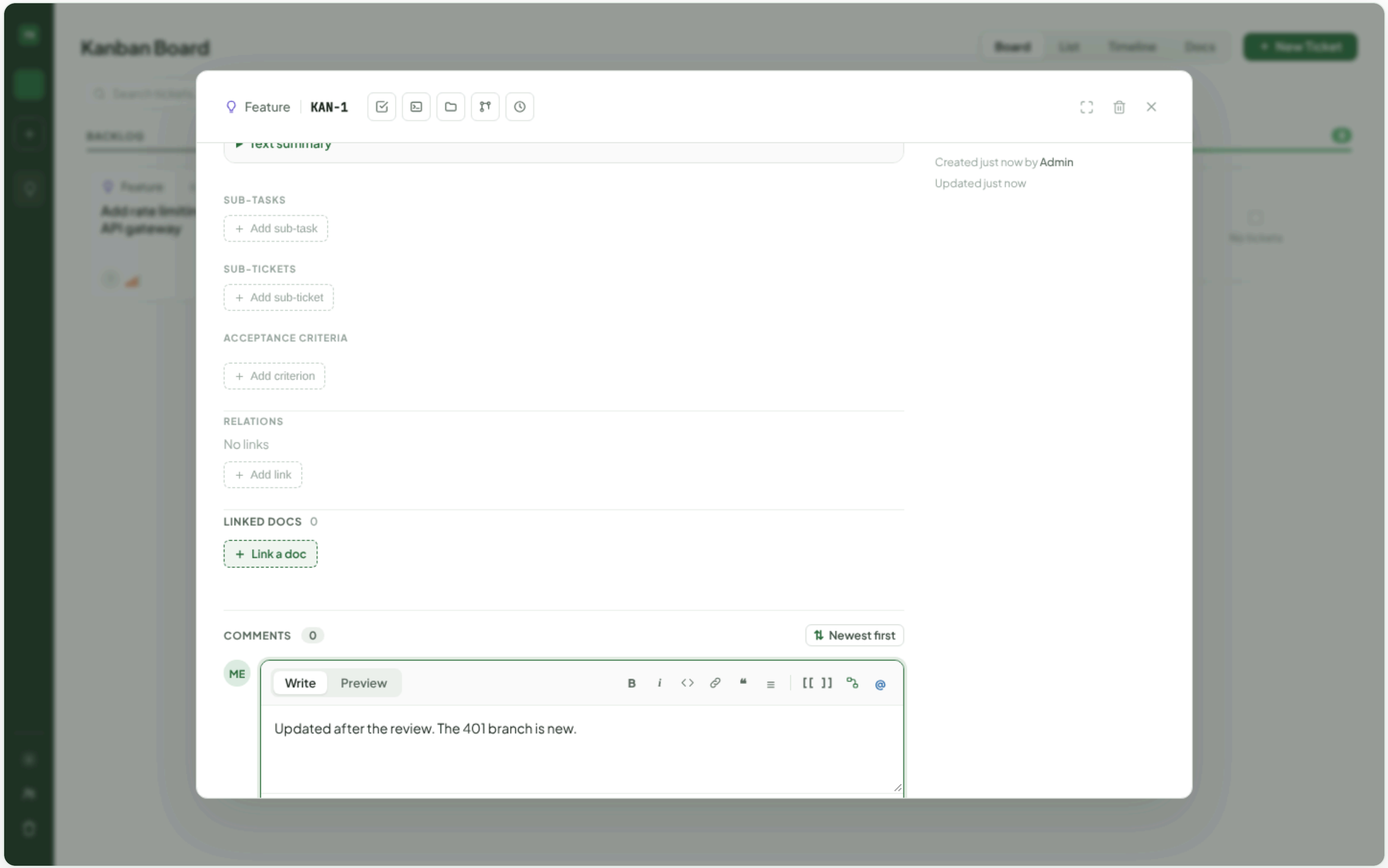
8. Full-size viewer

Scroll to zoom, drag to pan, +/- and 0 on the keyboard, Esc to close. Download is available here too.



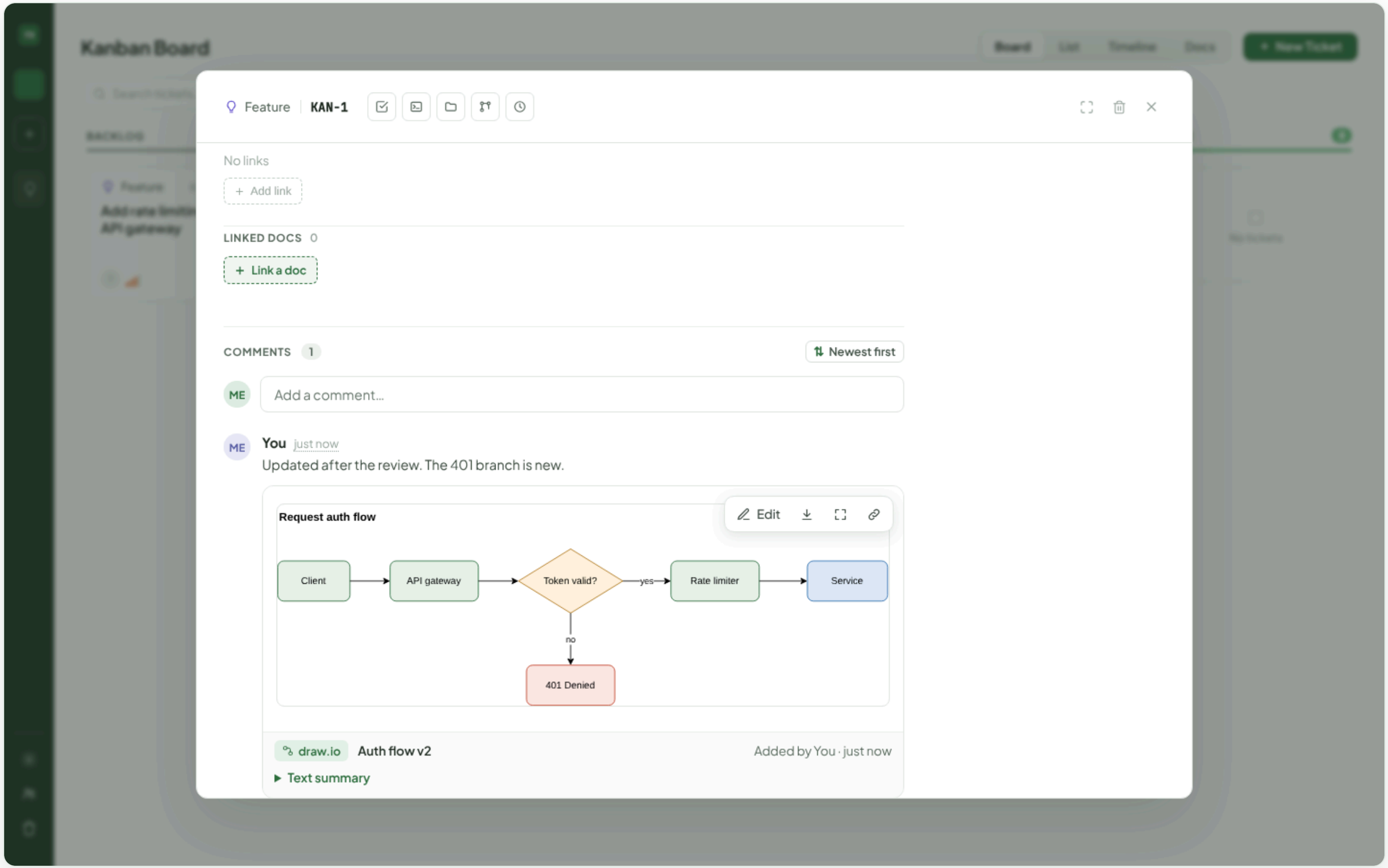
9. Diagrams in comments: the composer

Icon-only Diagram button next to the formatting tools (draw.io icon). The comment text is kept while the editor layer is open.



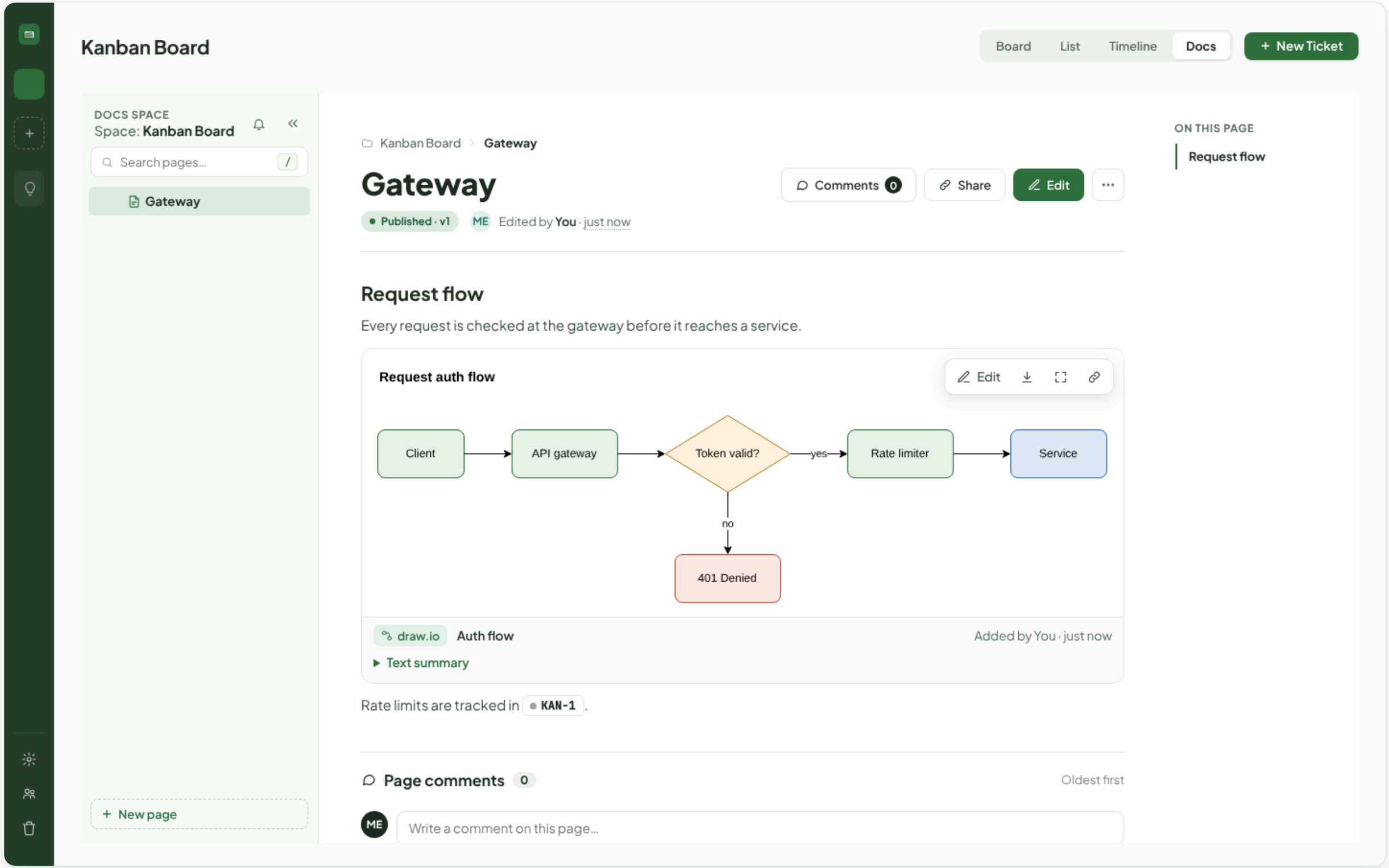
10. A comment with its diagram

Same actions as in the description, in a narrower layout.



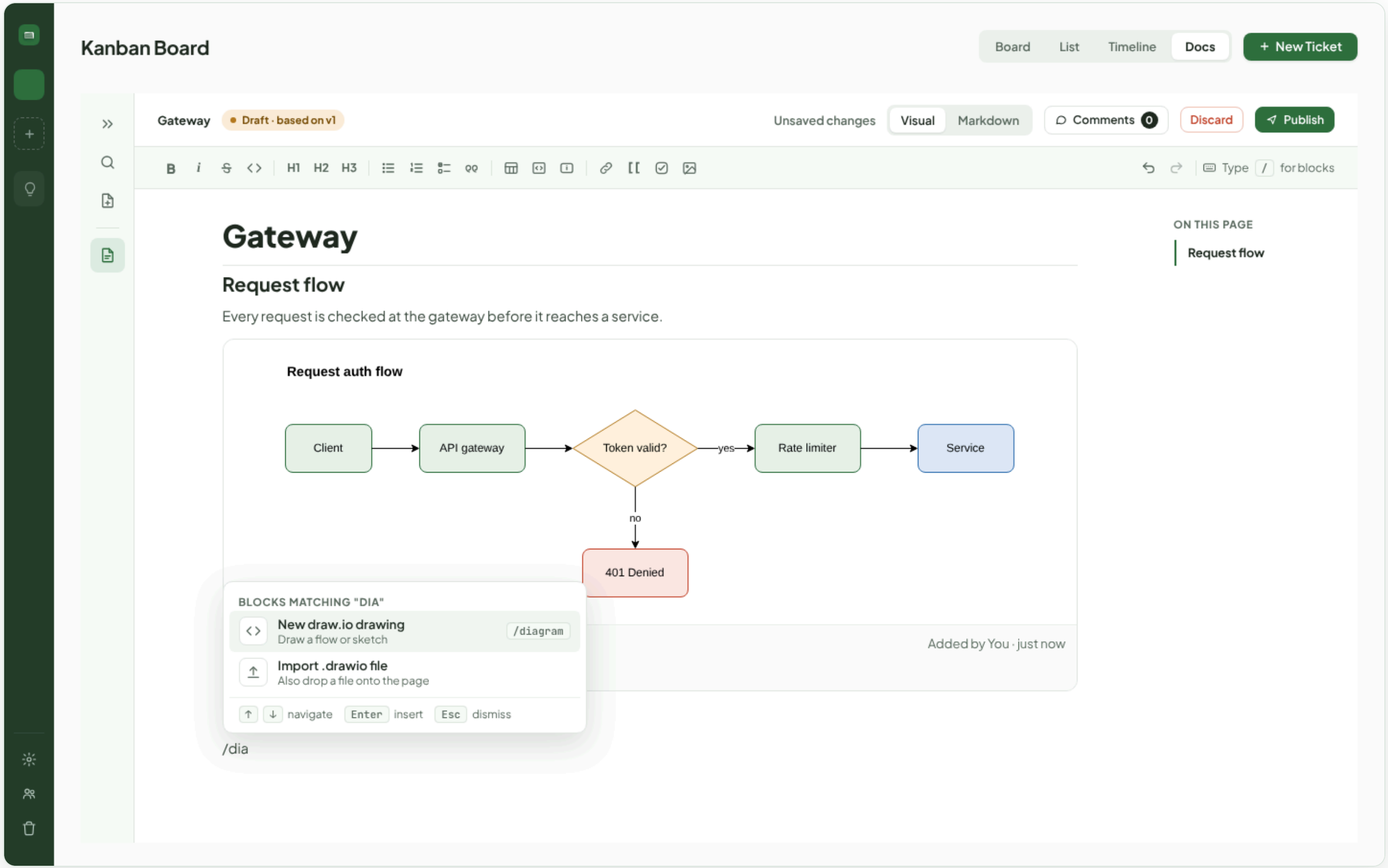
11. Diagrams in a Docs page

Wide layout in a published page, same hover toolbar.



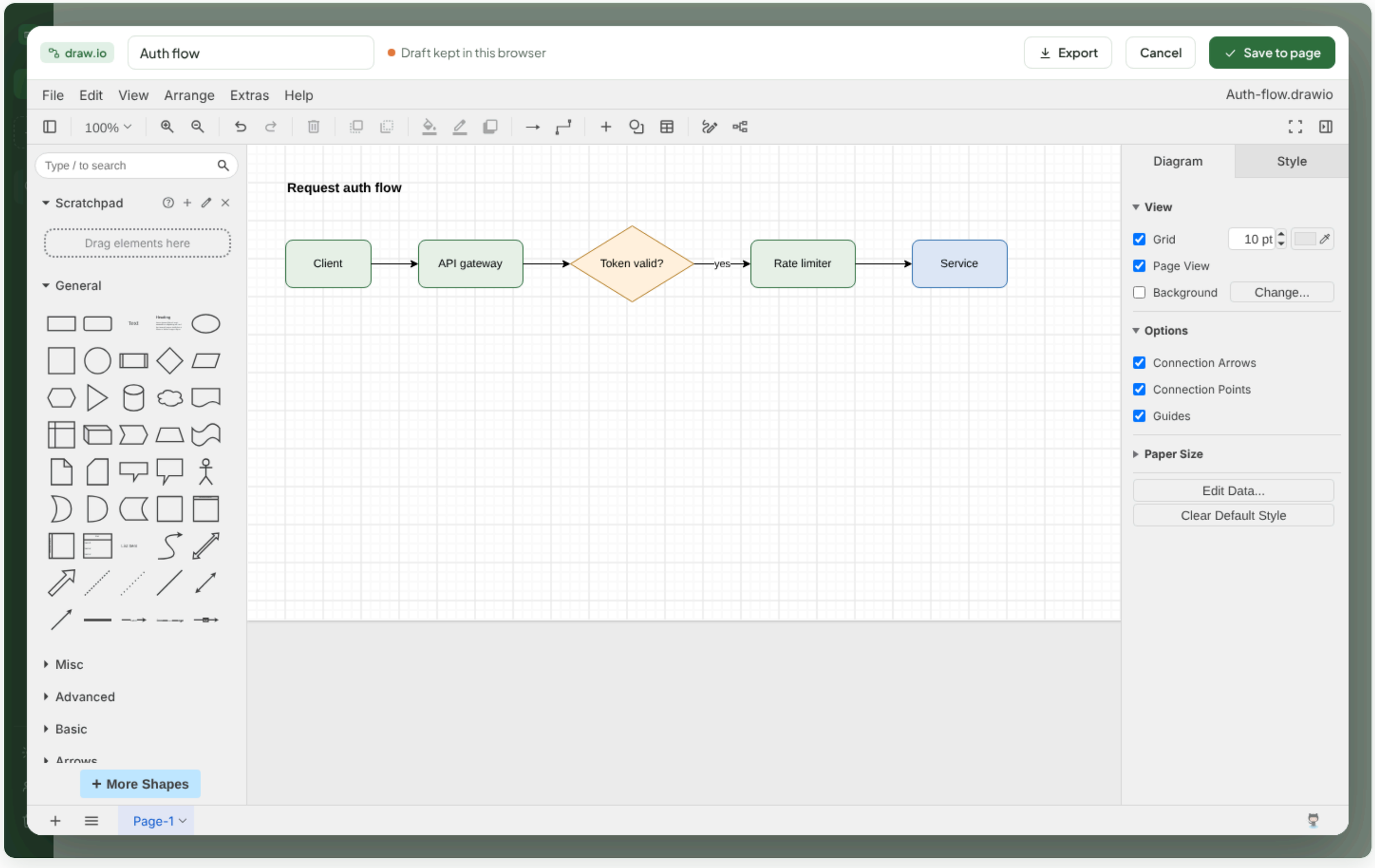
12. /diagram slash command in Docs

Typing /dia offers "New draw.io drawing" and "Import .drawio file" (a file can also be dropped onto the page).



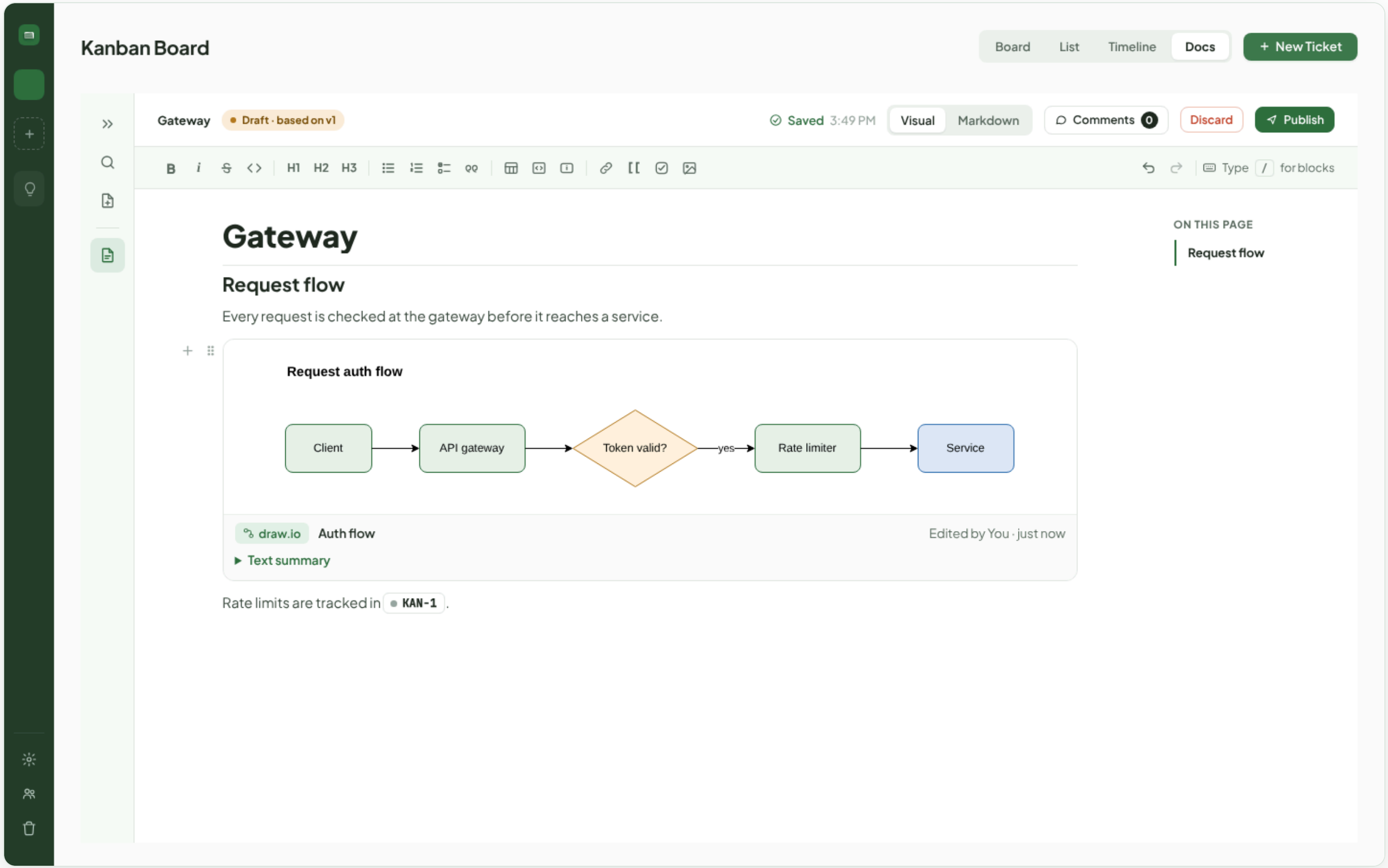
13. Editing a diagram that is already on a page

Edit opens the saved .drawio in the editor. Here the 401 Denied box is deleted; the header shows the local draft.



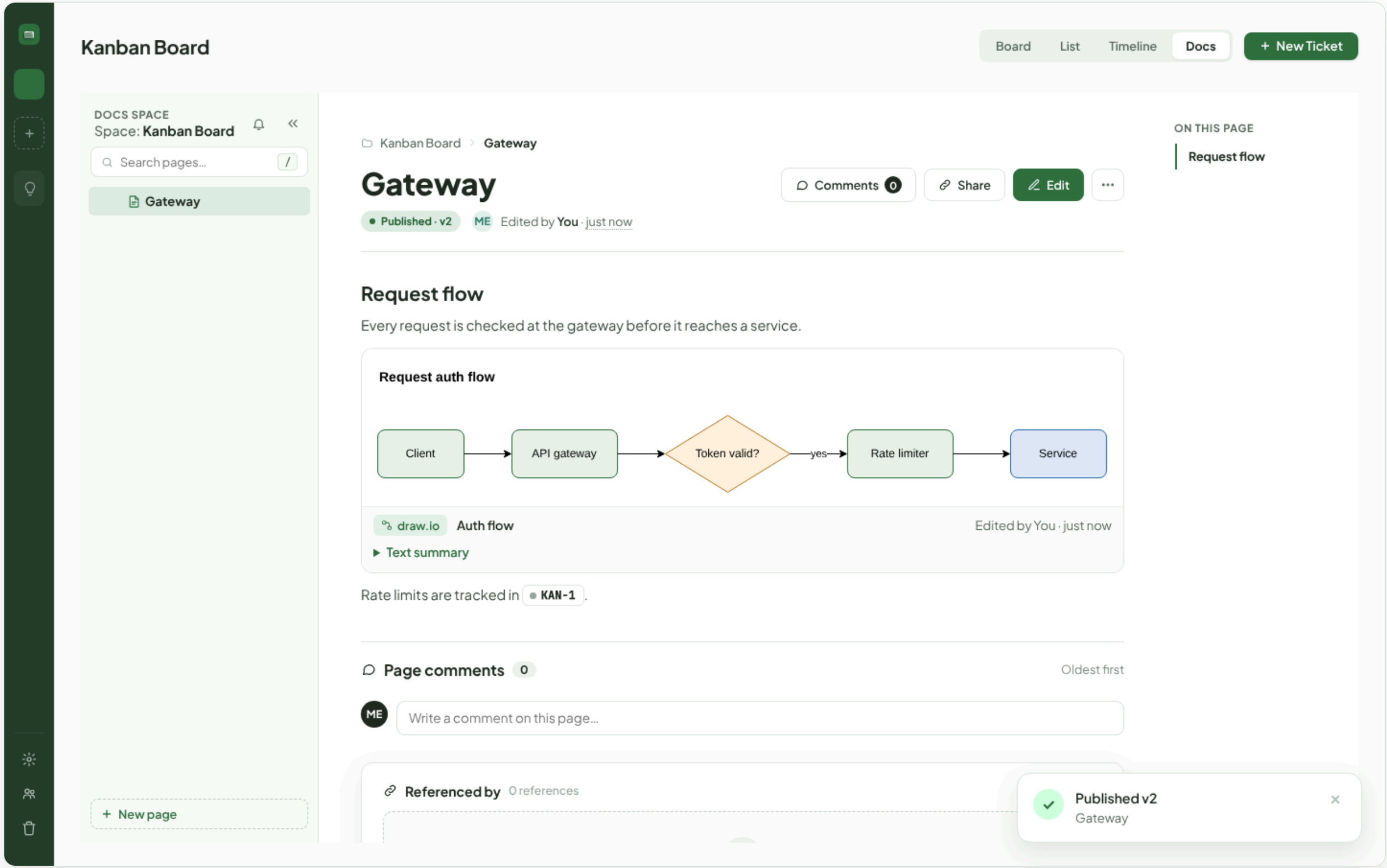
14. Saved: the page shows the new version

The diagram is edited in place (version 2). The page text did not change; the line still points at the same diagram.



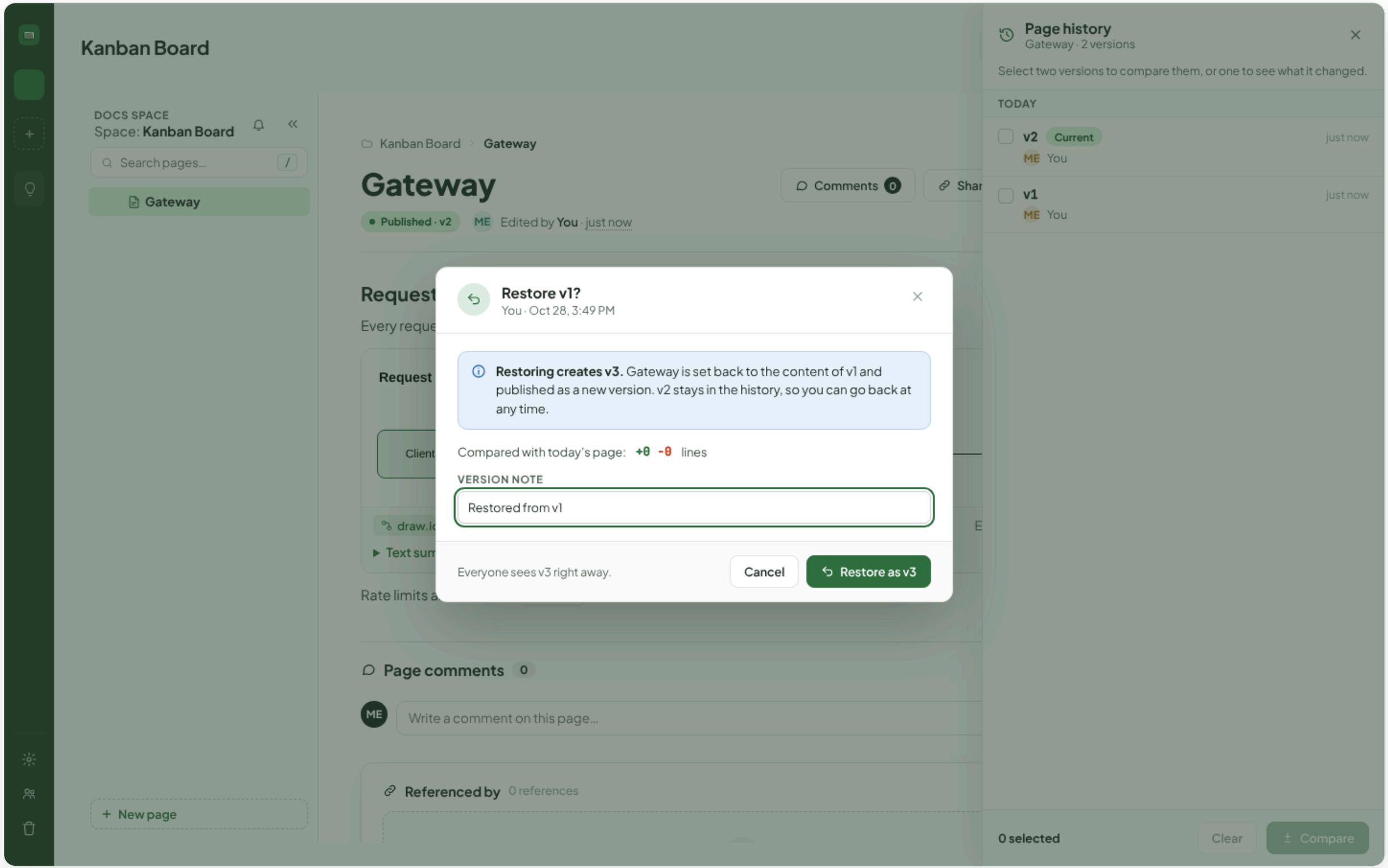
15. Publishing pins the diagram version

Page v2 is published. Each page version records the diagram versions it used.



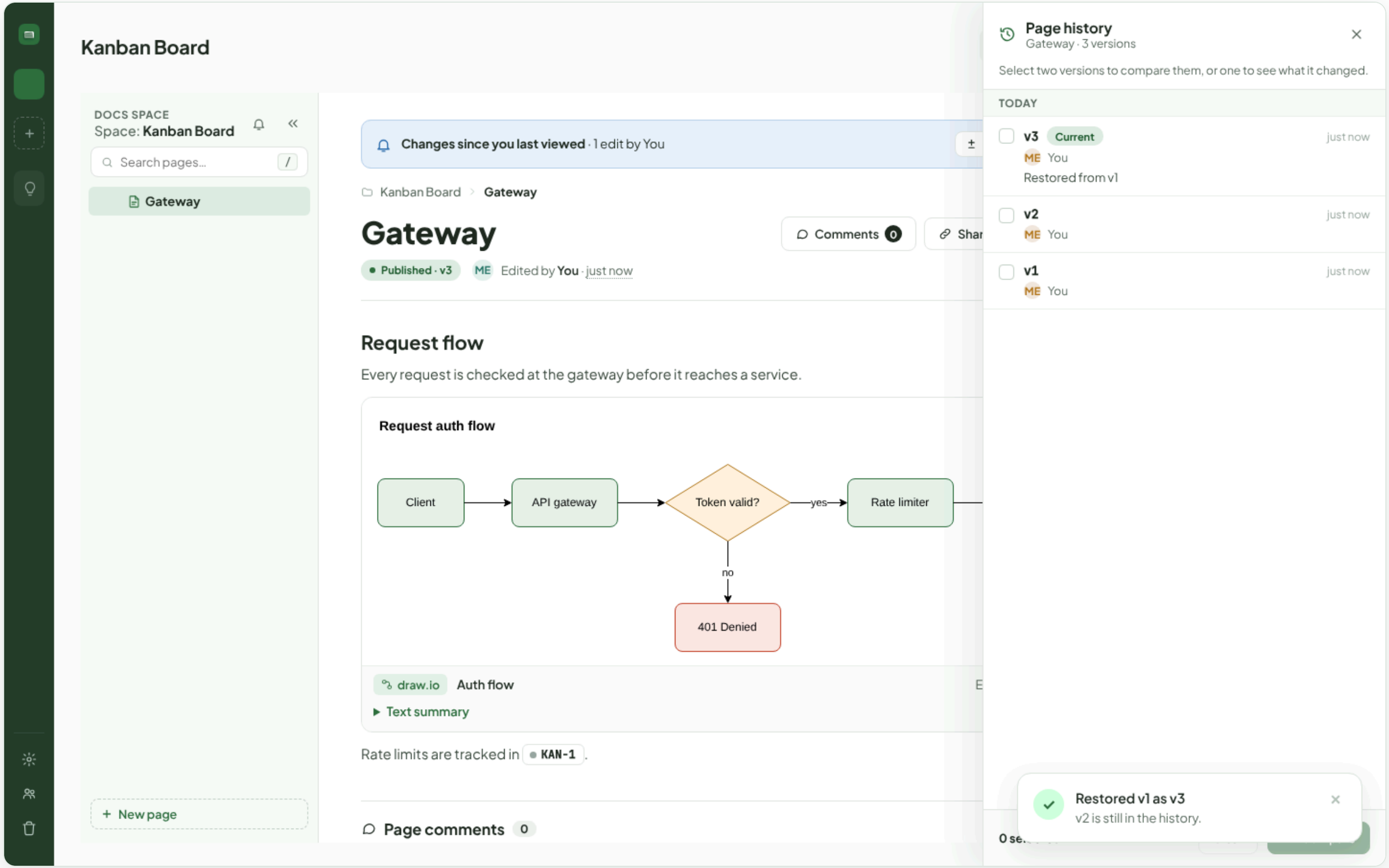
16. Restore an older page version

Page history › v1 › Restore.



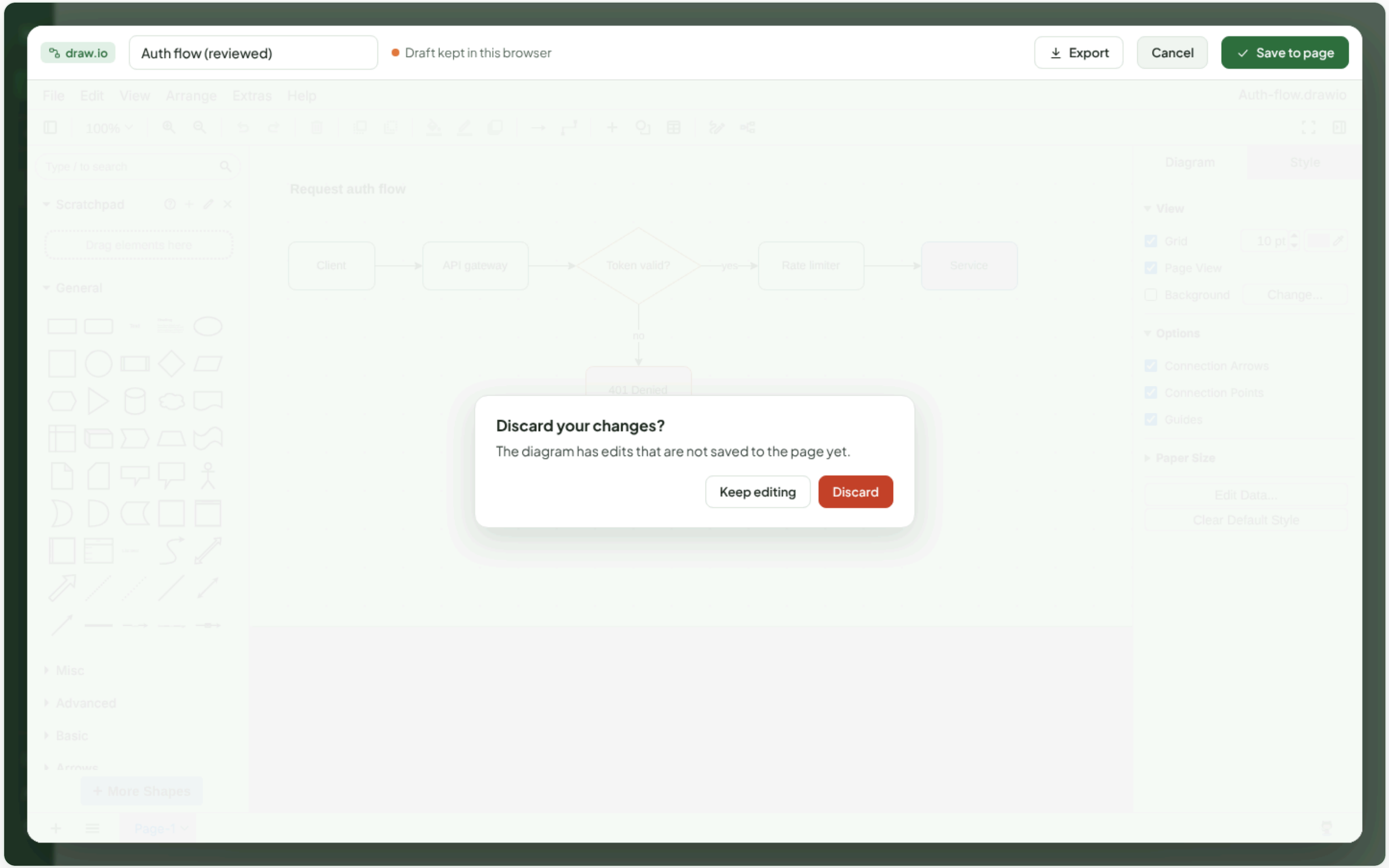
17. Restoring the page restored its diagram

v3 is v1 again and the diagram has its 401 Denied branch back (stored as a new diagram version, so nothing is lost).



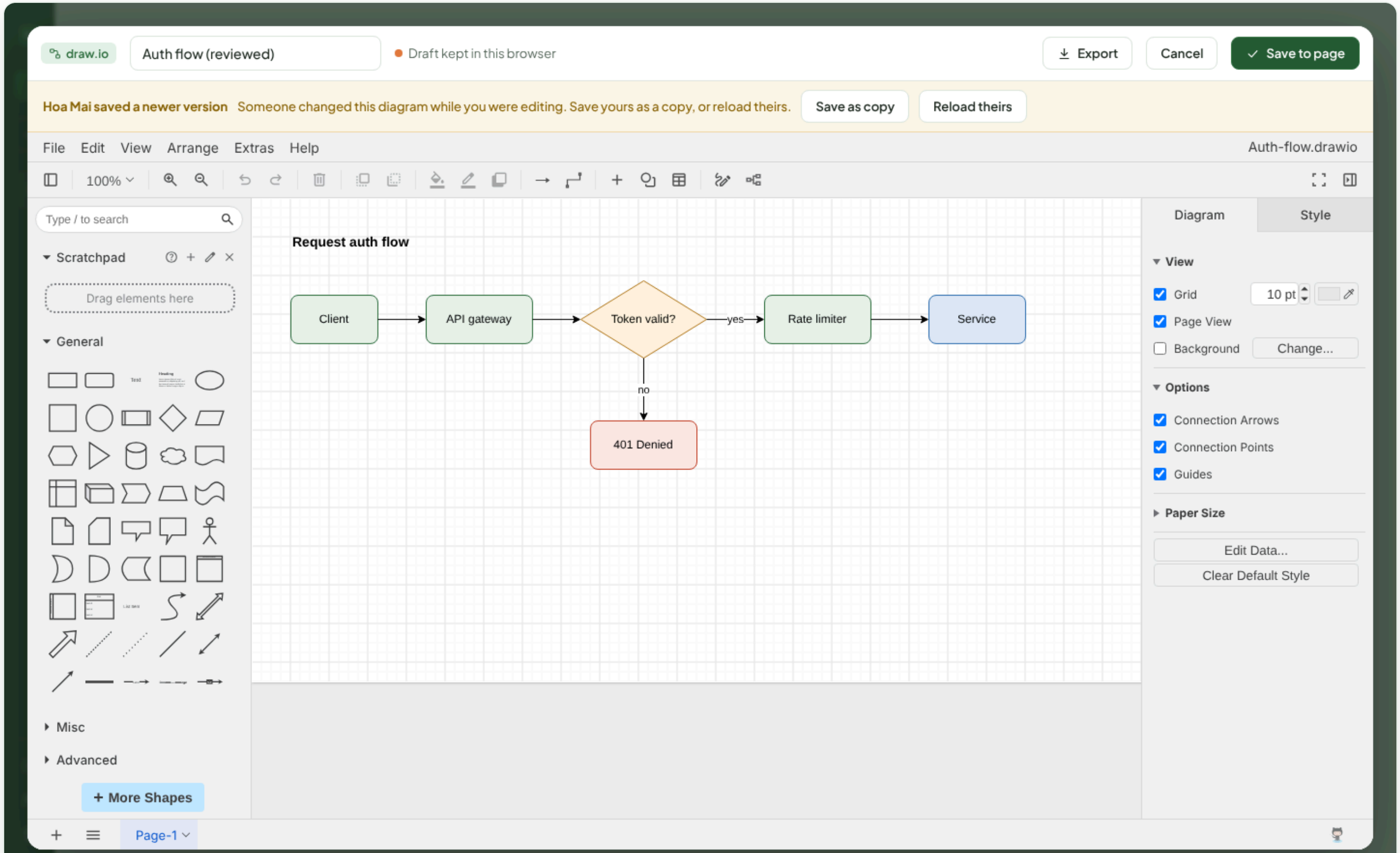
18. Closing the editor with unsaved edits

"Discard your changes?" with Keep editing / Discard.



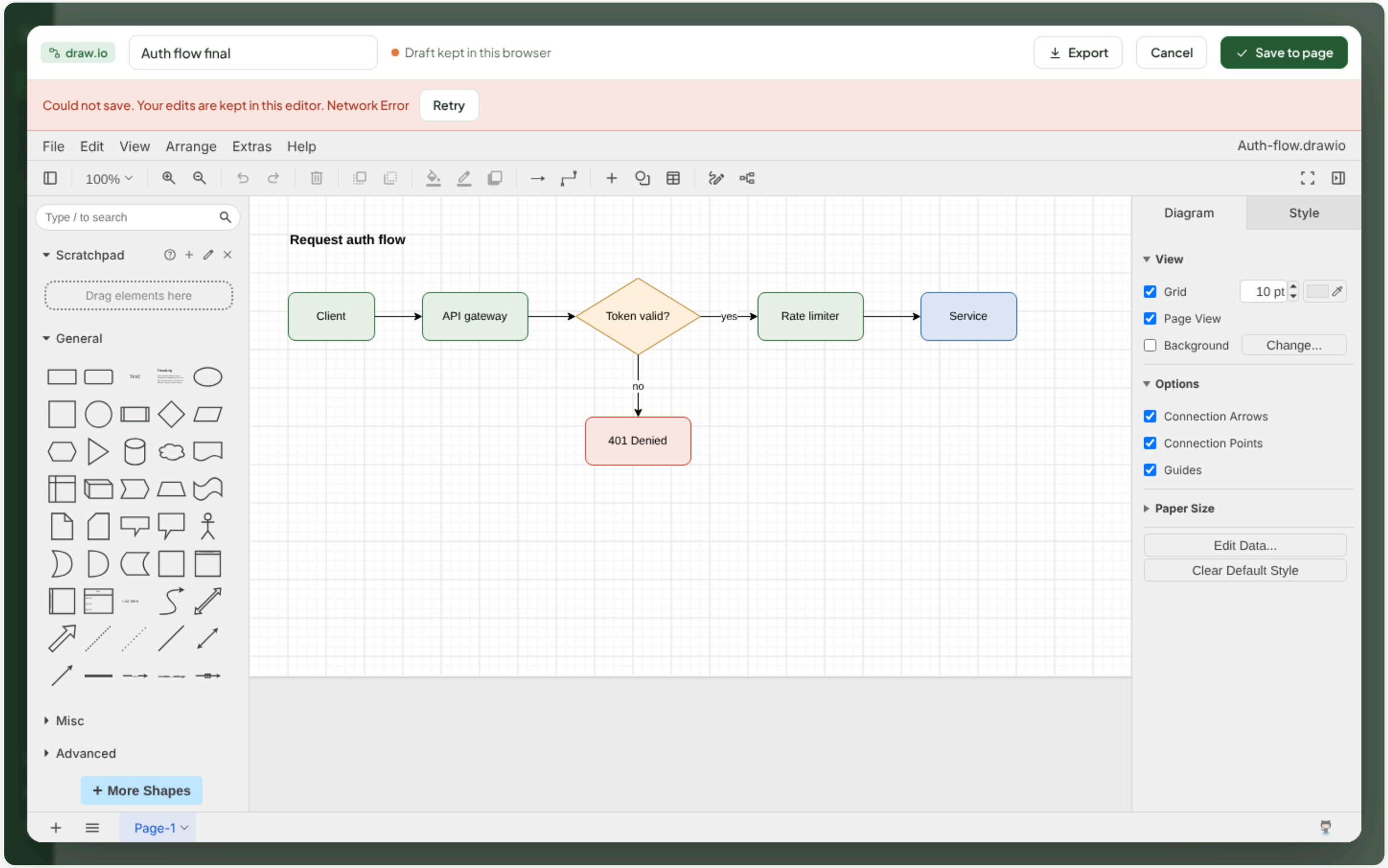
19. Edit conflict

Hoa Mai saved a newer version while this editor was open. Save as copy keeps your drawing as a new diagram; Reload theirs loads the newer one. (Server: 409 diagram_conflict.)



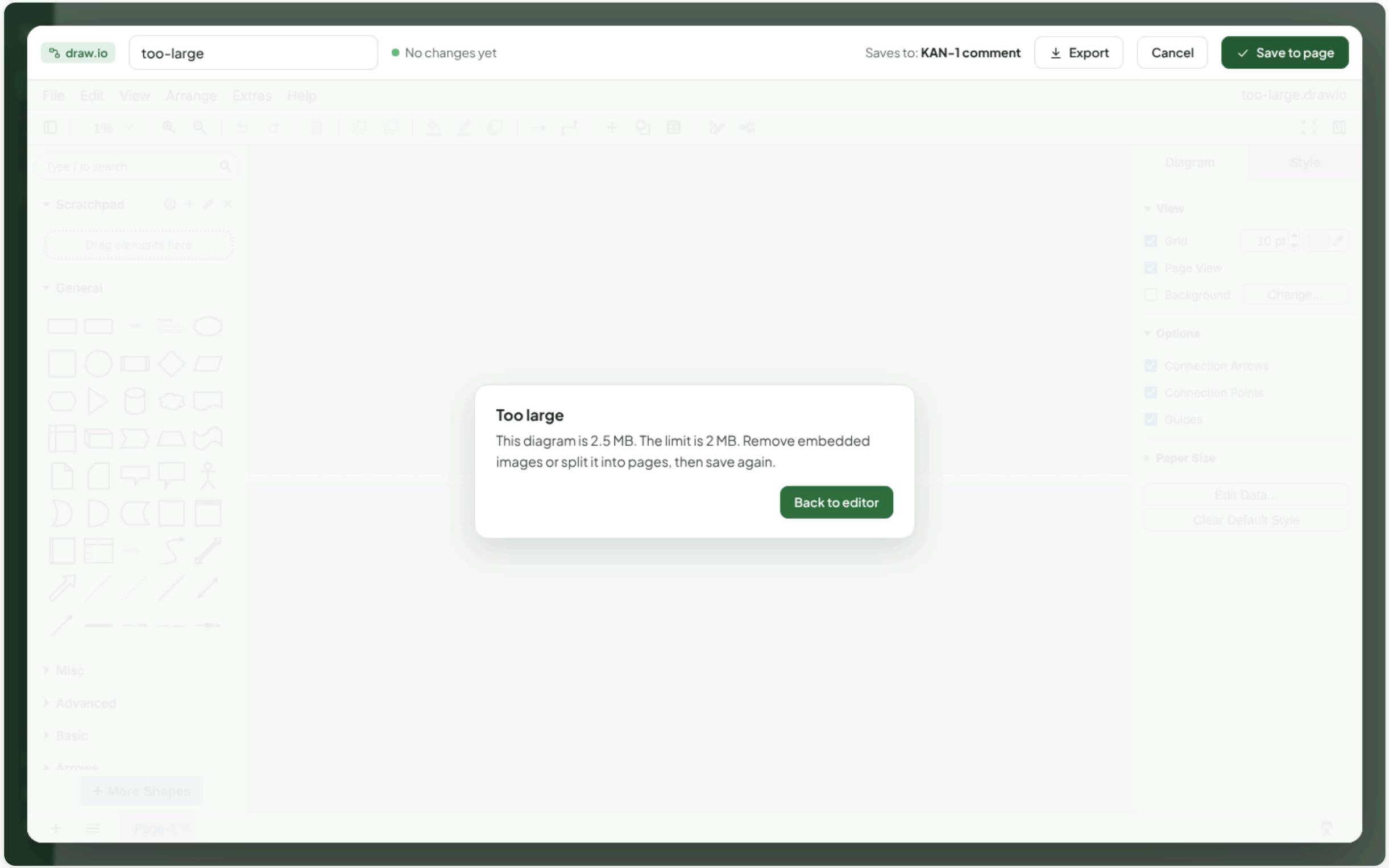
20. Save failed

The request failed; the edits stay in the editor and Retry is offered.



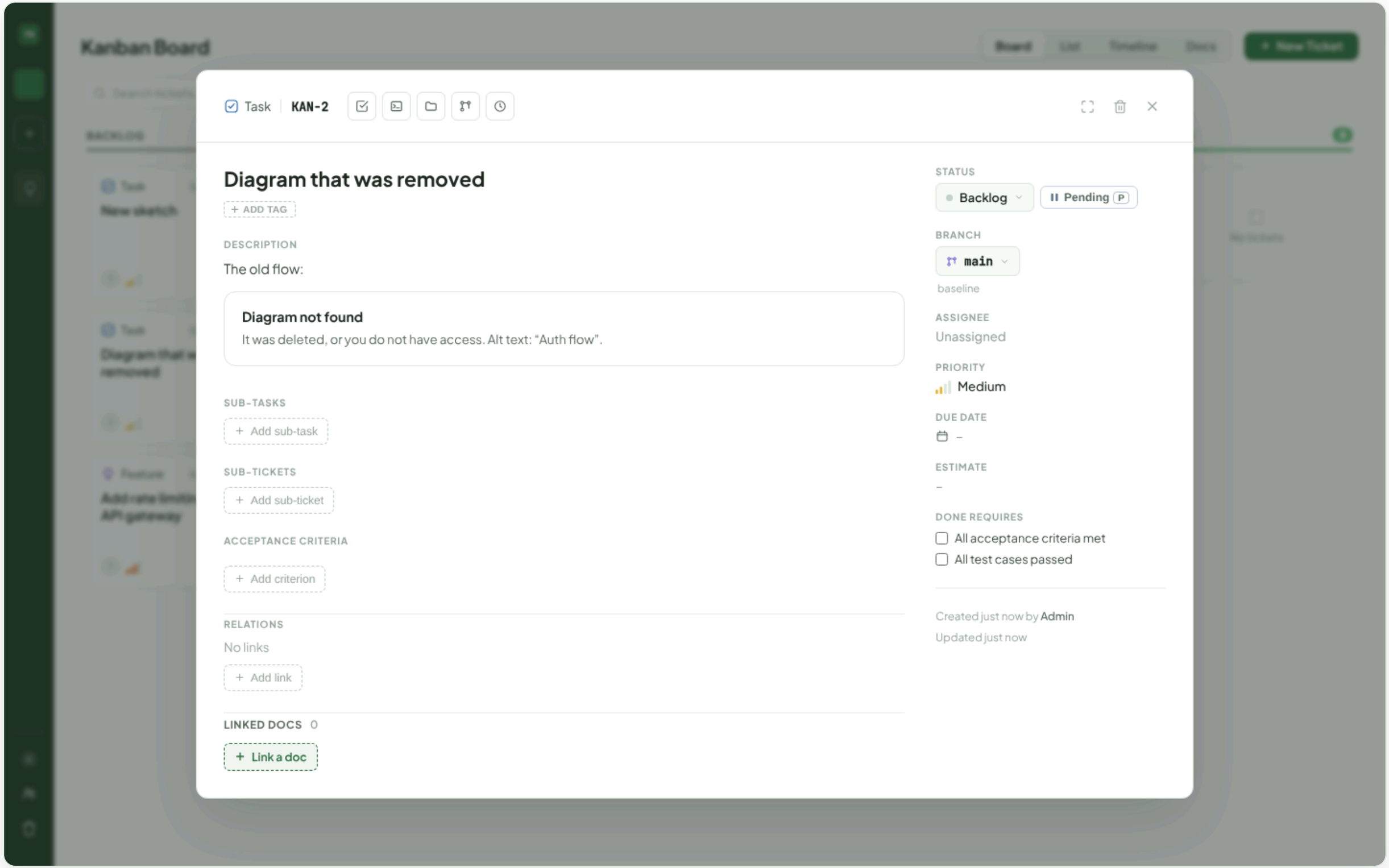
21. Too large

A 2.5 MB diagram is refused with a clear message (limit 2 MB per file; 413 diagram_too_large) and Back to editor.



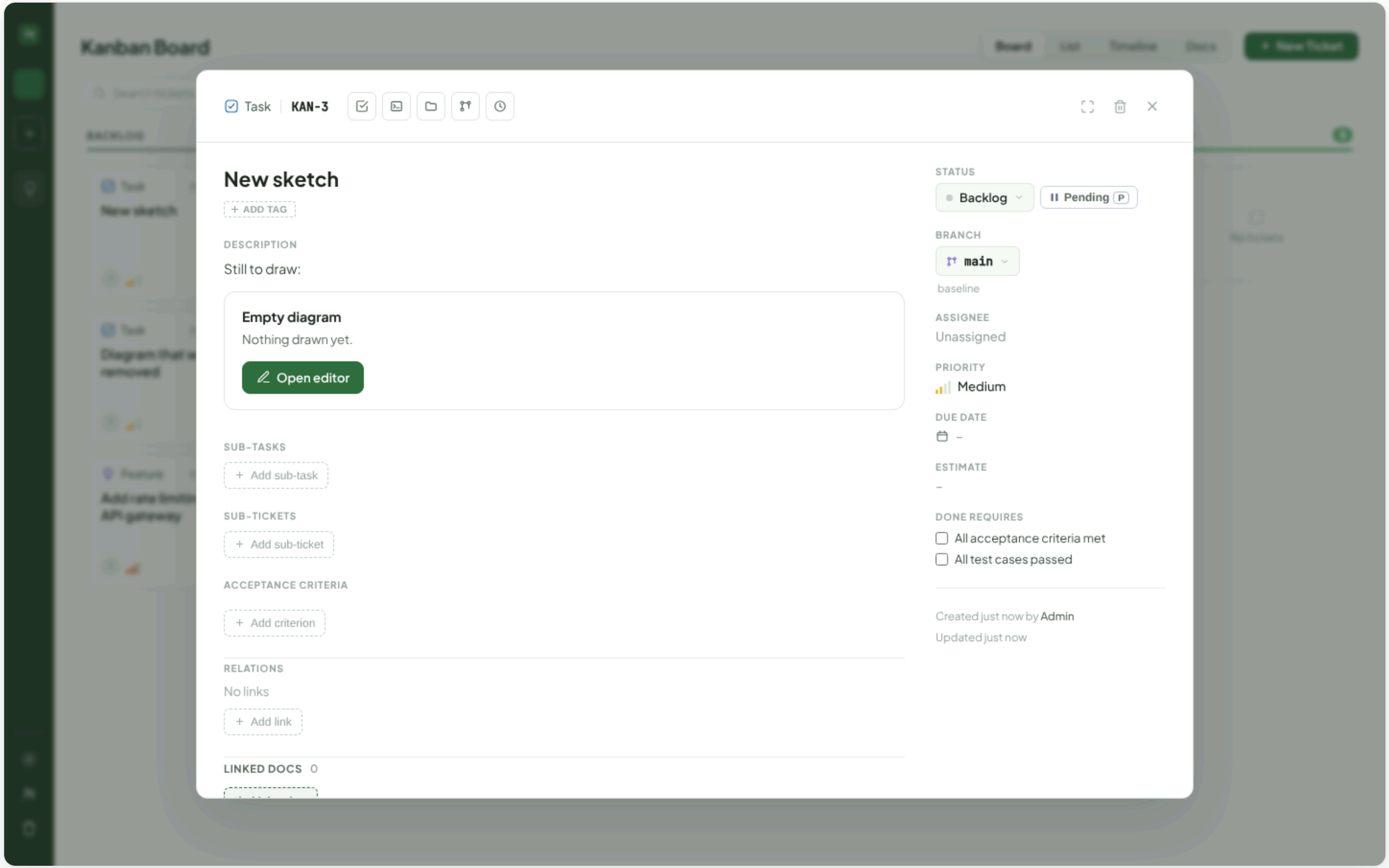
22. Missing diagram

A line that points at a deleted diagram (or one you cannot access) shows "Diagram not found" with the alt text.



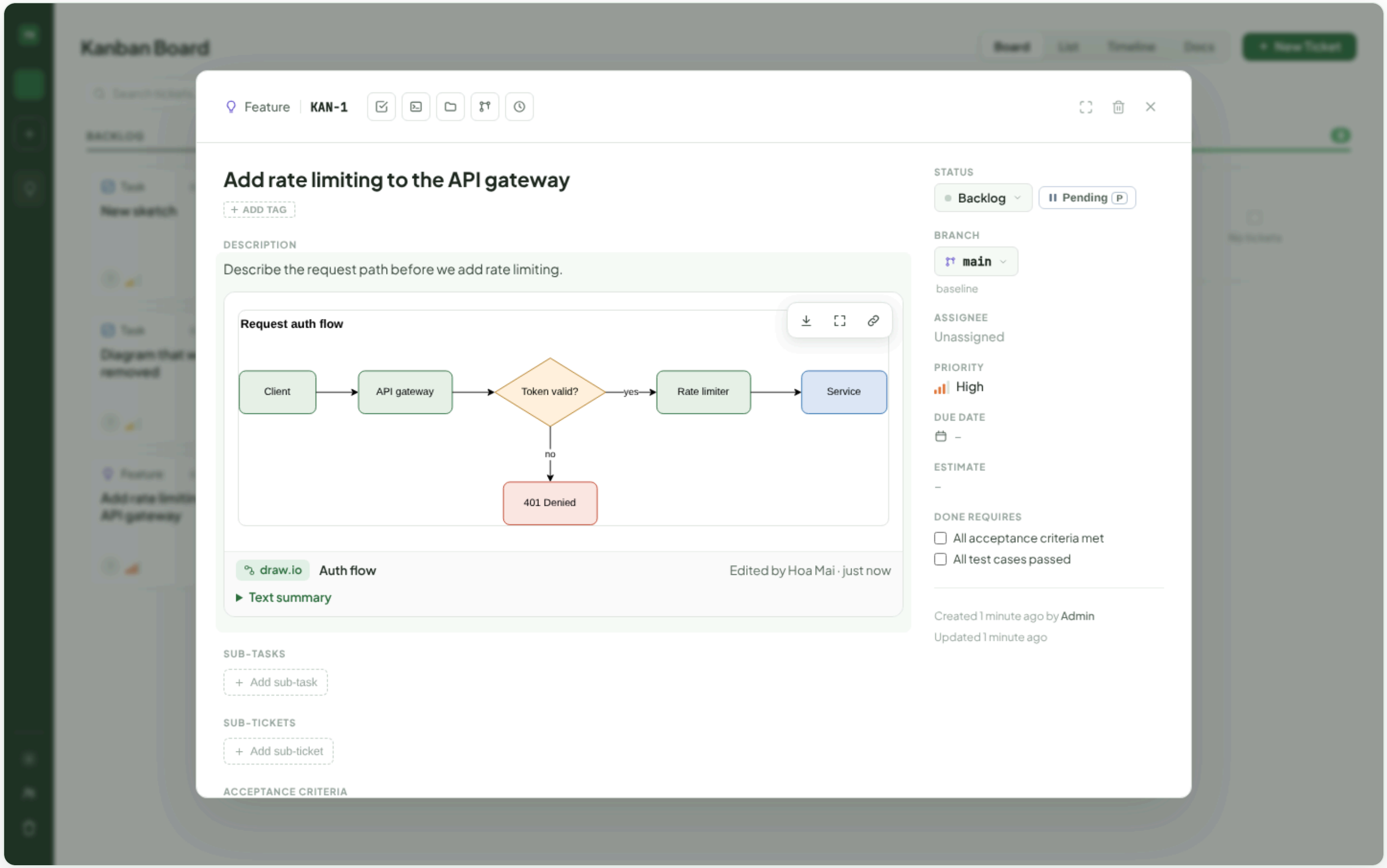
23. Empty diagram

"Nothing drawn yet" with Open editor.



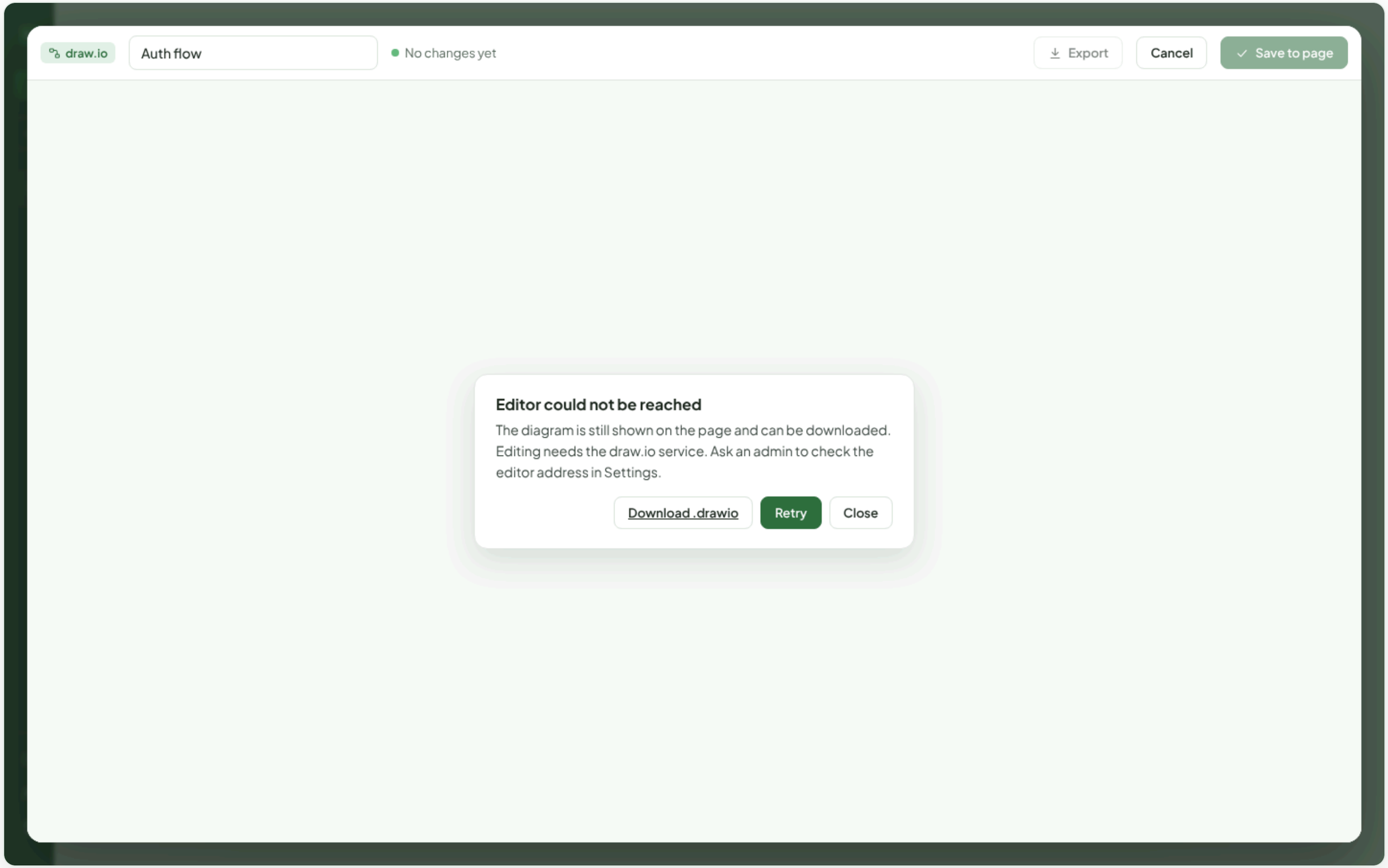
24. Read-only: viewers look and download

Signed in as a viewer: no Edit button; download, full size and copy link remain. Server returns 403 read_only for any write.



25. Editor unreachable

Settings pointed at a dead address. After the timeout: "Editor could not be reached" with Retry and Download .drawio. The diagram itself is still shown on the page and downloadable.



What was run

Backend	ruff check · ruff format --check · pytest: 428 passed (13 new in tests/test_diagrams.py)
Frontend	prettier --check · eslint (0 errors, 40 warnings = the existing baseline, none in new files) · tsc + vite build
Real browser	Chromium driven by Playwright against the real server + UI and the real draw.io webapp (jgraph/drawio, served from disk): every screen in this document is a screenshot of that run
Migration	alembic upgrade head on an empty database creates diagram, diagram_version, diagram_settings and docs_version.diagram_versions

Not in this version: phones are view, zoom and download only (no editing); the "New ticket" form has no Diagram action yet (the description toolbar appears once the ticket exists).